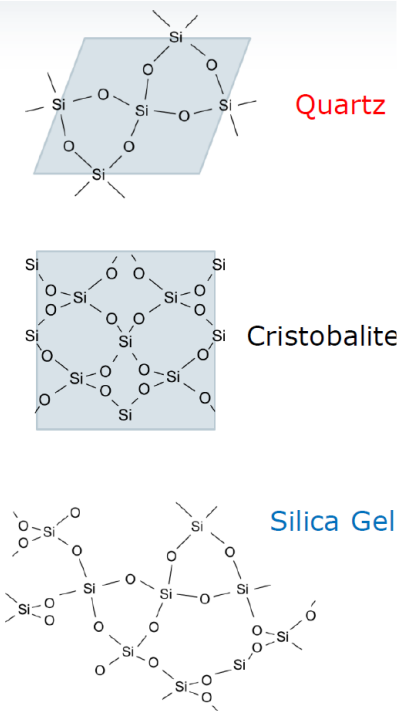
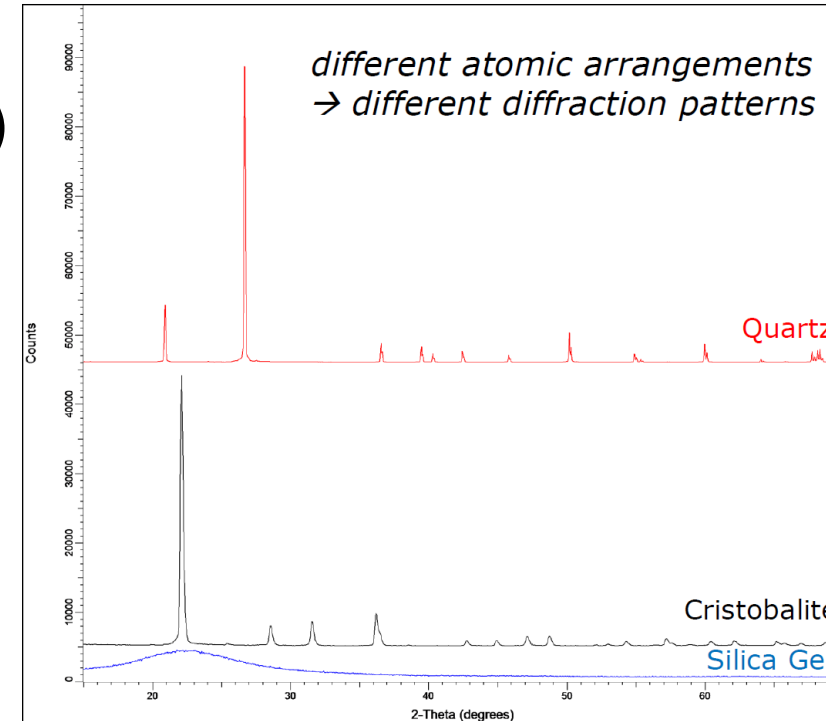


Diffraction analyses

- Phase identifications (crystalline and amorphous)
- Crystal structure determination
- Crystal structure refinements
- Texture analysis
- Residual stress analysis
- Quantitative phase analysis (and crystallinity determination)
- Microstructural analyses (crystallite sizes - microstrain)
- Order-disorder transitions and compositional analyses
- Thin films



Calculating a diffraction pattern

- The **structure factor** sums the result of scattering from all of the atoms in the unit cell (f_0) to form a diffraction peak from the (hkl) planes of atoms.
- The intensity of Bragg peak is proportional to F_{hkl}^2 .

$$I_{hkl} \propto |F_{hkl}|^2 \qquad F_{hkl} = N_k \sum_{n=1}^N f_0 e^{2\pi i(hx_n + ky_n + lz_n)} \qquad f_0: \text{Fourier transform of } \rho(r)$$

x_n, y_n, z_n : position of atoms in the atomic planes (where the atoms are on the atomic planes)

f_0 : **atomic scattering factor** (fattore di forma atomico). It depends on nature of atom, direction of scattering, and X-ray wavelength and quantifies the relative efficiency of scattering at any angle by the group of electrons in each atom

$$f_0 \rightarrow Z \text{ for } \theta \rightarrow 0^\circ$$

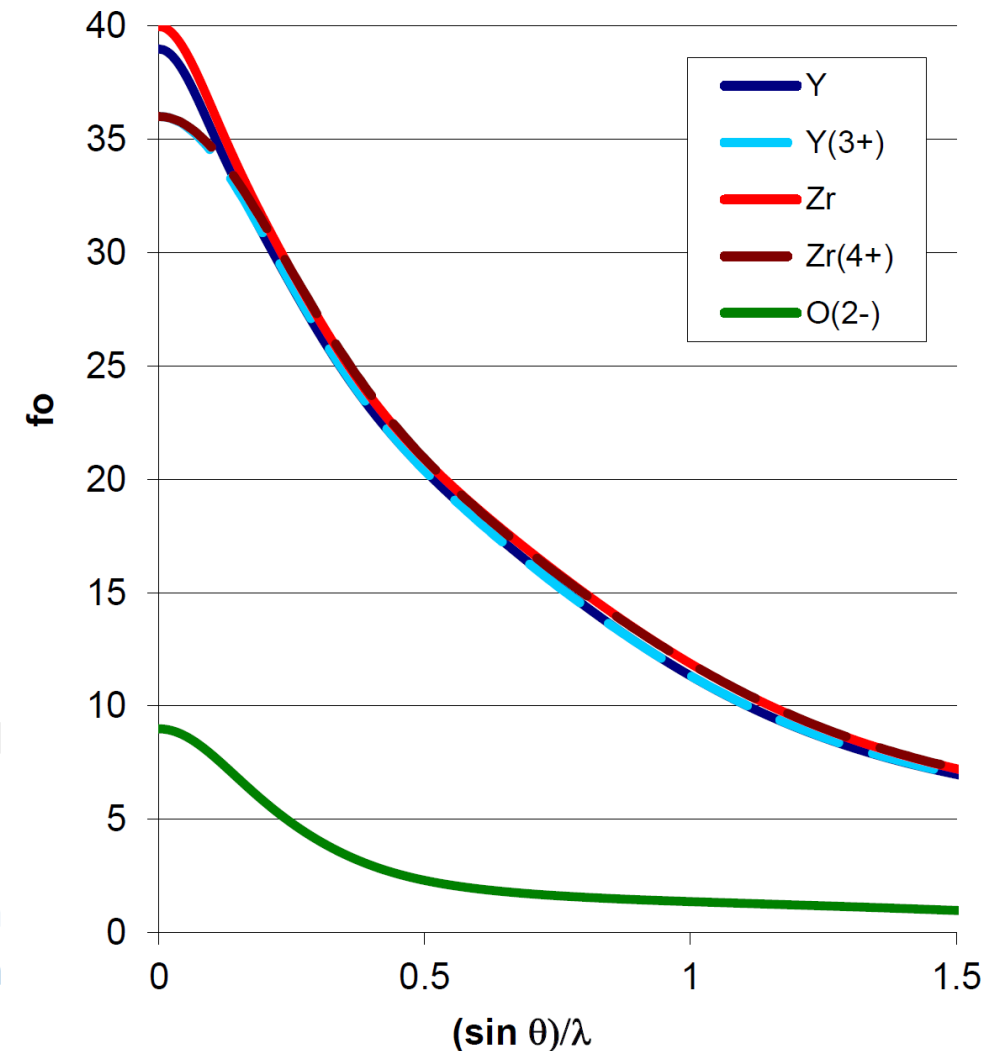
N_k : fraction of every equivalent position occupied by atom k. Often an atom does not occupy every unit cell of the crystallite. This is the probability of the atom occupying position xyz

Atomic scattering factor

$$f_0 = \frac{\text{Amplitude scattered by } Z \text{ electrons in an atom}}{\text{Amplitude scattered by a single electron}}$$

The atomic scattering factor f_0 can be found using tables or equations determined experimentally or from quantum mechanical approximations

- f_0 at 0° is equal to the number of electrons around the atom
- the variation with $(\sin \theta)/\lambda$ depends on size of atom
 - larger atoms drop off quicker
- The number of electrons around an atom defines how strongly it will scatter the incident X-ray beam
- f_0 represents the scattering by an atom at 0 Kelvin. Changes in temperature affect the thermal motion of atoms, and this in turn affects the scattered intensities.



Thermal motion (Debye-Waller, B)

- the efficiency of scattering by an atom is reduced because the atom and its electrons are not stationary (the atom is vibrating about its equilibrium lattice site)
- the amount of vibration is quantified by the Debye-Waller temperature factor B:

$$B = 8\pi^2 U^2$$

U^2 is the mean-square amplitude of the vibration

- This factor only reduces the intensity of the peaks and does not change the sharpness or shape of the peaks.

$$f = f_0 e^{\left[-\frac{B \sin^2 \theta}{\lambda^2} \right]}$$

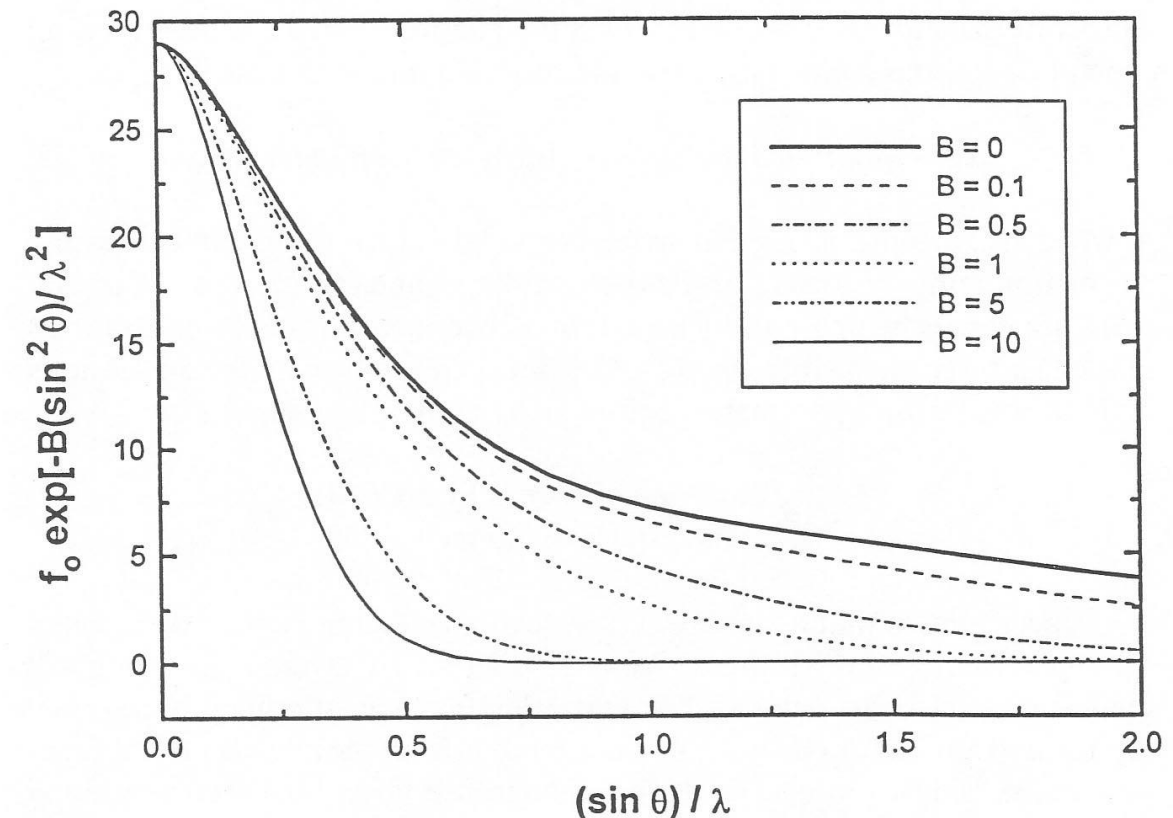


Figure 3.14. The effect of atomic thermal motion on the copper scattering factor.

Simulation of diffraction pattern

Ideal powder diffraction patterns can be simulated if you know ...

- space group symmetry
- unit cell dimensions
- atom types
- relative coordinates of atoms in unit cell
- atomic site occupancies
- atomic thermal displacement parameters
- ...

for each and every phase in the sample

Crystallographic Information File (CIF):

file structure based on tag–value ASCII pairs with tags defined in machine-readable dictionaries.

Powder diffraction file (PDF):
file

Where to Get Crystal Structures?

- Inorganic Crystal Structure Database (ICSD)
- CCDC Cambridge Structure Database (CSD) (organic materials) (<http://beta-www.ccdc.cam.ac.uk/pages/Home.aspx>)
- Crystallography Open Database (COD) (<http://www.crystallography.net/>)

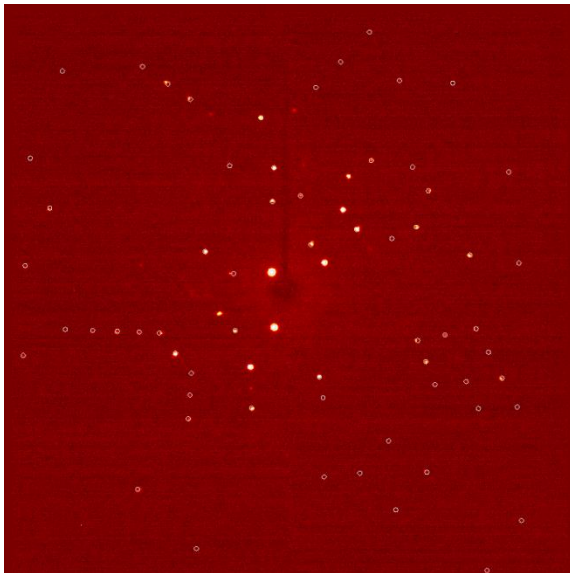
Formula Al2 O3									
Name Aluminum Oxide									
Name (mineral) Corundum, syn									
Name (common)									
Lattice: Rhombo, H Axes			Mol. weight = 101.96						
S.G.: R-3c (167)			Volume [CD] = 254.84						
			Dx =						
			Dm =						
a = 4.75920			I/lcor = 0.980						
c = 12.99200									
a/b = 1.00000									
c/b = 2.72987									
Z = 6									
Additional Patterns: See PDF 01-074-1081, 01-075-0782, 01-083-2080, 01-081-2267 and 01-075-1862									
Deleted Or Rejected By: Delete: 00-010-0173 is experimental and as good, WFM 12/98									
General Comments: Calculation of diffractometer peak intensities done with MICRO-POWD v. 2.2 (D. Smith and K. Smith) using default instrument broadening function (NBS Table), diffracted beam monochromator polarization correction, and atomic scattering factors corrected for anomalous dispersion. Cell parameters from D. Smith documentation for MICRO-POWD sample file (original structure data after Newnham and DeHaan)									
Atomic positions from same source: Al in 12c with z=0.352, O in 18e with x=0.306									
Isotropic thermal parameters also from Smith: Al, B=0.14; O, B=0.22									
Unit Cell Data Source: Powder Diffraction									
Structure									
Publication: Acta Crystallogr., Sec. A: Cryst. Phys., Diff., Theor. Crystallogr.									
Detail: volume 46, page 271 (1990)									
Primary Reference									
Publication: ICDD Grant-in-Aid									
Authors: Grier, D., McCarthy, G., North Dakota State University, Fargo, North Dakota, USA.									
Radiation: CuKα1			Filter: M						
Wavelength: 1.54060			d-spacing:						
h:									
SS/FOM: 428.5 (0.0023,30)									

Formula Na Cl		d	2θ	I fix	h	k	l
Name Sodium Chloride		3.26000	27.335	14	1	1	1
Name (mineral) Halite, syn		2.82100	31.693	101	2	0	0
Name (common)		1.99400	45.450	56	2	2	0
		1.70100	53.853	3	3	1	1
		1.62800	56.479	16	2	2	2
		1.41000	66.229	7	4	0	0
		1.29400	73.066	2	3	3	1
		1.26100	75.304	12	4	2	0
		1.15150	83.973	8	4	2	2
		1.08550	90.409	2	5	1	1
		0.99690	101.193	3	4	4	0
		0.95330	107.809	2	5	3	1
		0.94010	110.045	4	6	0	0
		0.89170	119.504	5	6	2	0
		0.86010	127.169	2	5	3	3
		0.85030	129.893	4	6	2	2
		0.81410	142.239	3	4	4	4
Additional Patterns: See PDF 01-072-1668 Color: Colorless General Comments: An ACS reagent grade sample recrystallized twice from hydrochloric acid Melting Point: 1077 K Temperature of Data Collection: Pattern taken at 299 K Unit Cell Data Source: Powder Diffraction							
Primary Reference Publication: Natl. Bur. Stand. (U. S.), Circ. 539 Detail: volume II, page 41 (1953) Authors: Swanson, Fuyat.							
Radiation: CuKα1	Filter: F d-spacing:						
Wavelength: 1.54060							
SS/FOM: 91.9 (0.0109,17)							

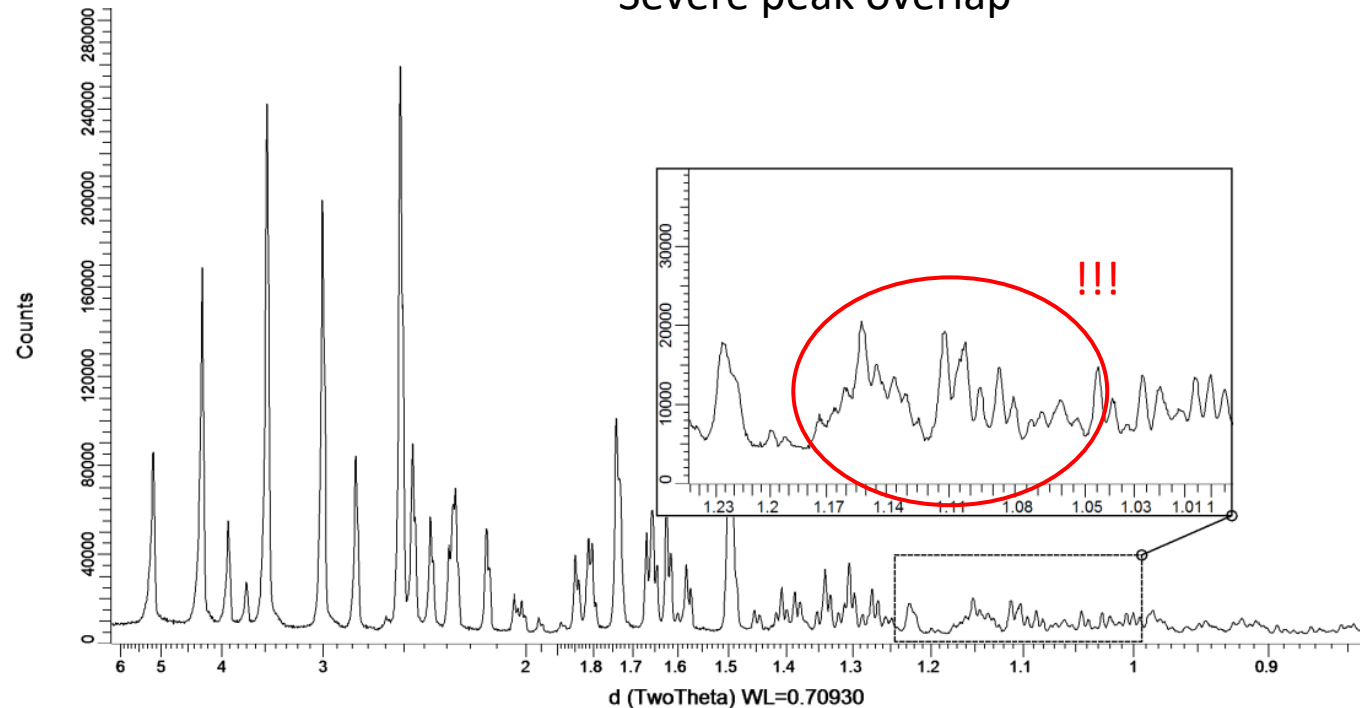
Introduction to Rietveld refinements: Rietveld's idea

- A diffraction pattern consists of multiple peaks (reflections) with intensities $I(hkl)$.
- To extract crystallographic information, these intensities $I(hkl)$ have to be determined.

Single crystal diffraction
Well-separated peaks



Powder diffraction
Severe peak overlap



Introduction to Rietveld refinements: Rietveld's idea

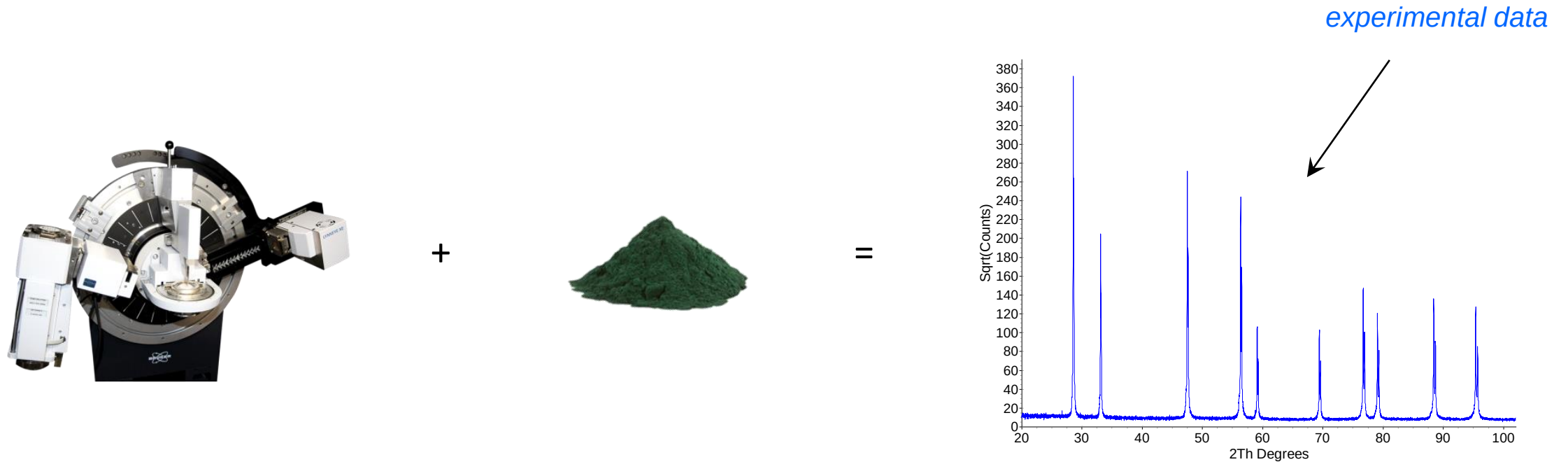
- The area of **well-separated peaks** could be determined easily (cut out of paper and weighted or later by computational methods)
- But overlapping reflections were a problem – and there were many of them, too!
- **Hugo M. Rietveld** (1966/'67/'69) realized that the observed peaks could be partitioned at every data point according to the relative contributions of the **overlapping reflections**.
- Further, all data points in a powder pattern could be described using **analytical and/or numerical functions**.
- The calculated curve could be adapted to the measured curve using a non-linear **least squares methods**

$$WSS = \sum_i w_i (I_i^{exp} - I_i^{calc})^2 \rightarrow \min$$

$$w_i = \frac{1}{\sqrt{I_i^{exp}}}$$

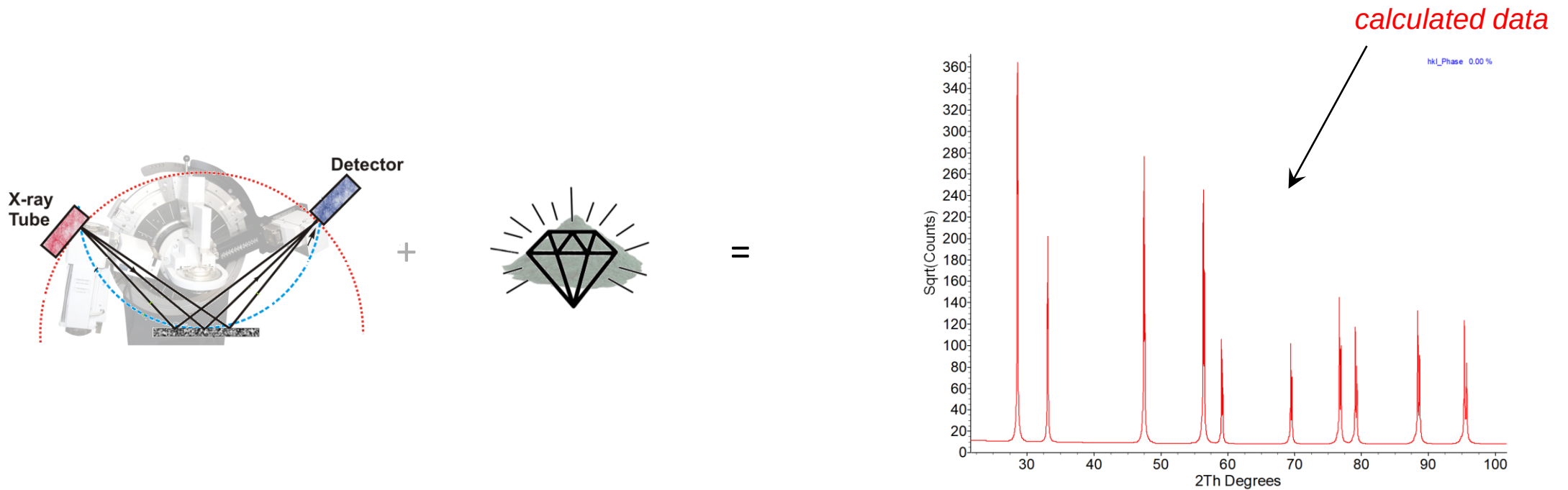
Weighted Sum of Squares

Introduction to Rietveld refinements: Background



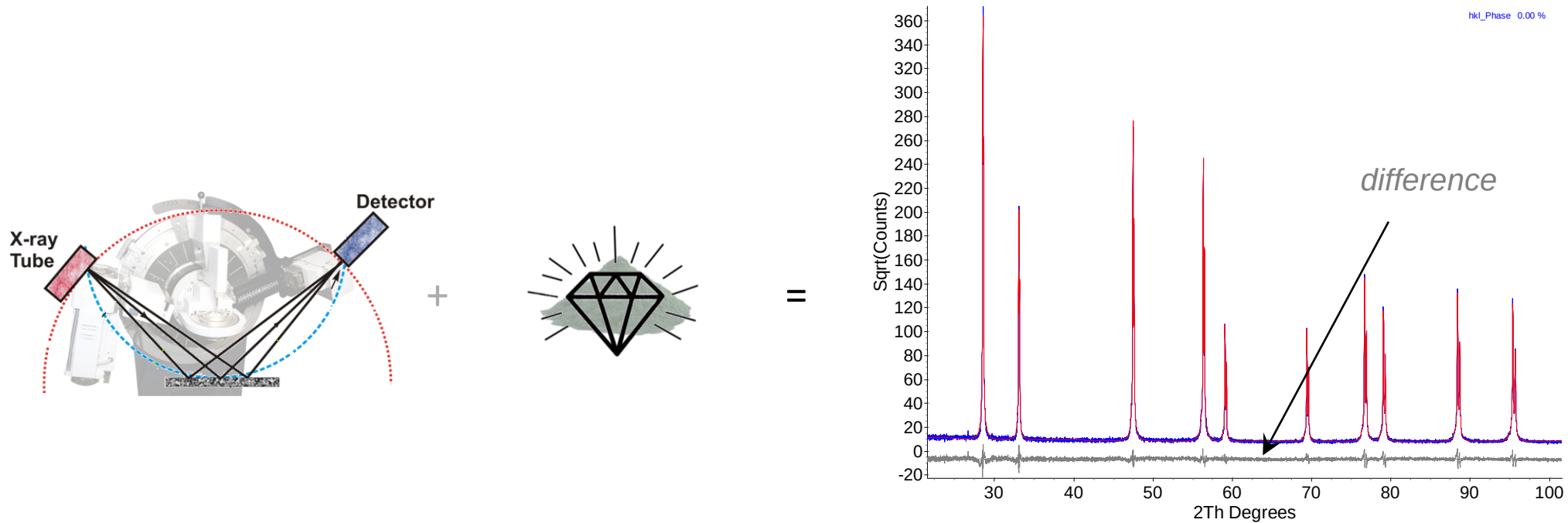
- In the XRD experiment we **measure a diffraction pattern**.
- This pattern is very specific. It depends on:
 - the **instrument** used
 - and the **sample**

Introduction to Rietveld refinements: Background



- In a Rietveld refinement, we build a **model** for both the **diffractometer** and the **sample**
- We can **calculate** the **diffraction pattern** of our model.
- And we can adjust our model with different **refineable parameters**.

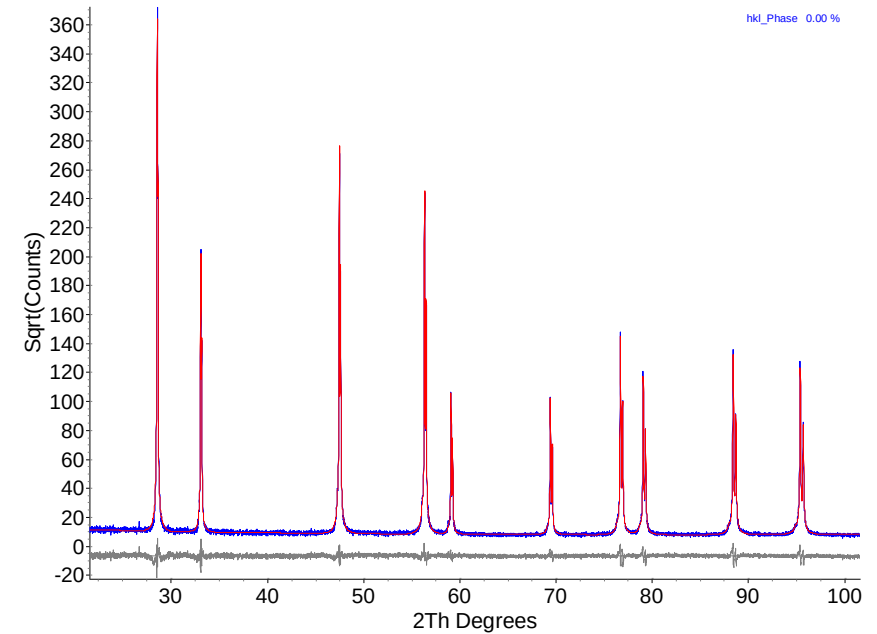
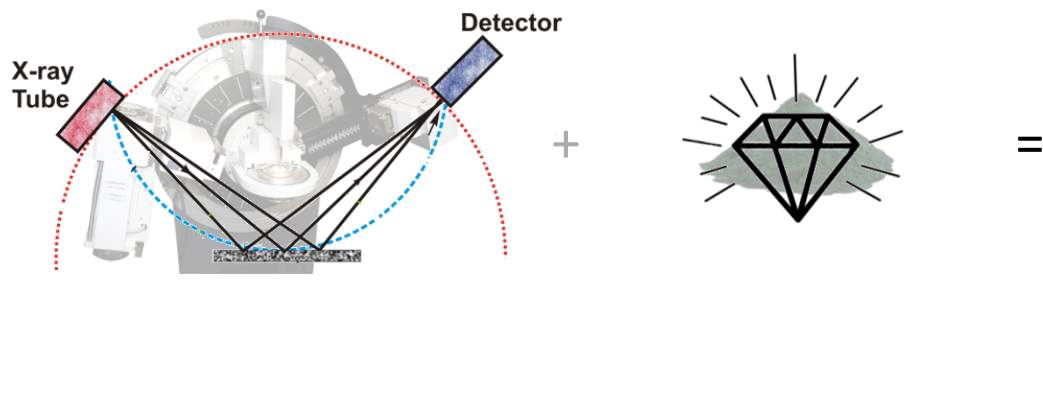
Introduction to Rietveld refinements: Background



With Rietveld software, such as MAUD, we **refine** the model:

- The value of a parameter is varied, so that the **difference** between the measured and the calculated diffraction profile is minimized.
- The **smaller** the **difference**, the **better** (=closer to the real sample) is our **model** for the sample.

Introduction to Rietveld refinements: What is the aim?



- The aim is to find a model, that is a close description of the sample.
- The **information on the sample** that can be extracted include:
 - Lattice parameters; atomic coordinates
 - Crystallite size, strain
 - Phase compositions

Introduction to Rietveld Method: Calculating diffraction patterns

➤ Calculation of a diffraction pattern includes:

- Crystal structures of phases present
- Contributions from the instrument

$$I_i^{calc} = S_F \sum_{j=1}^{N \text{ phases}} \frac{f_j}{V_j^2} \sum_{k=1}^{N \text{ peaks}} L_k |F_{k,j}|^2 S_j(2\theta_i - 2\theta_{k,j}) P_{k,j} A_j + bkg_i$$

$$|F_{k,j}|^2 = N_k \left| \sum_{n=1}^N f_n e^{-B_n \frac{\sin^2 \theta}{\lambda^2}} (e^{2\pi i(hx_n + ky_n + lz_n)}) \right|^2$$

S_F : Scale factor;

L : Lorentz-Polarization factor

F^2 : Structure factor (with multiplicity)

S : Profile shape function

P : Preferred orientation

A : absorption factor

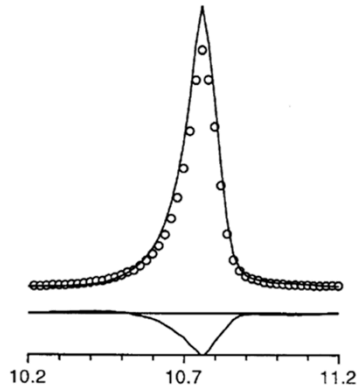
bkg : Background

Introduction to Rietveld Method: Calculating peak intensities

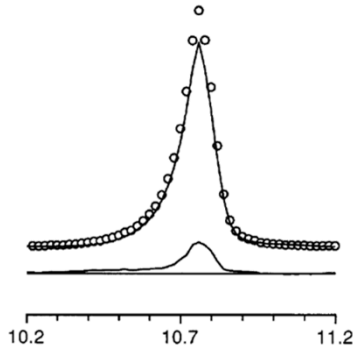
$$I_i^{calc} = S_F \sum_{k=1}^{N \text{ peaks}} L_k |F_k|^2 S(2\theta_i - 2\theta_k) A_j P_k + bkg_i$$

- **S_F**: scale factor of each phase combines different experimental parameters (i.e. size of the sample, intensity of the primary beam, influence of slit widths)
- **L**: Lorentz-Polarization factor (instrument geometry, monocromator, detector, beam size, angular sample positioning)
- **F**: structure factor (multiplicity of the k reflection, temperature factor B, atomic scattering factor)
- **P**: texture or preferred orientation
- **S(2θ_i-2θ)**: profile shape function (Gaussian, Voigt, pseudo-Voigt, etc.)
- **A**: absorption factor (depends on the absorption coefficient of the sample)
- **bkg**: background (polynomial function in 2θ)

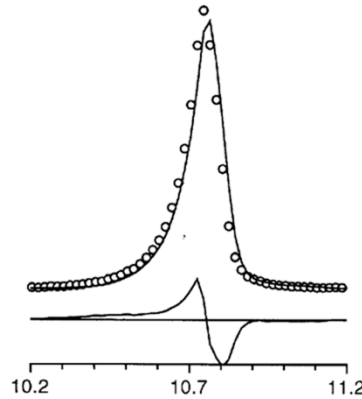
Visual inspection



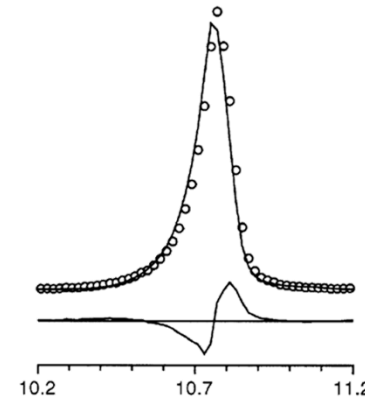
Intensity too high



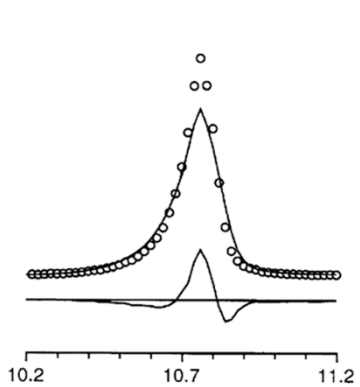
Intensity too low



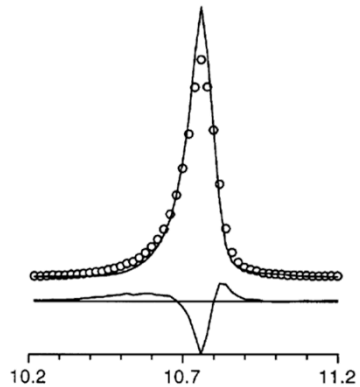
Position too far right



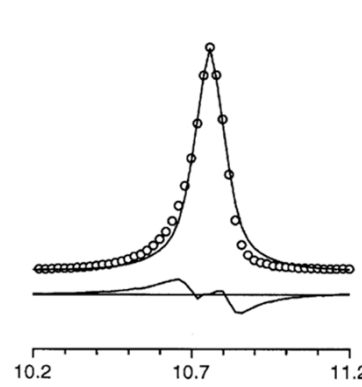
Position too far left



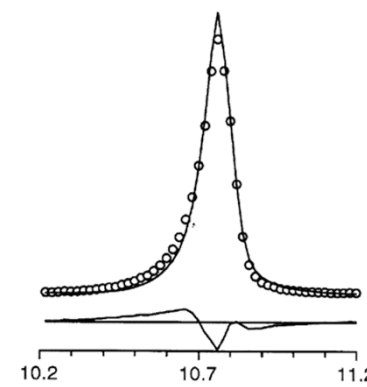
FWHM too large



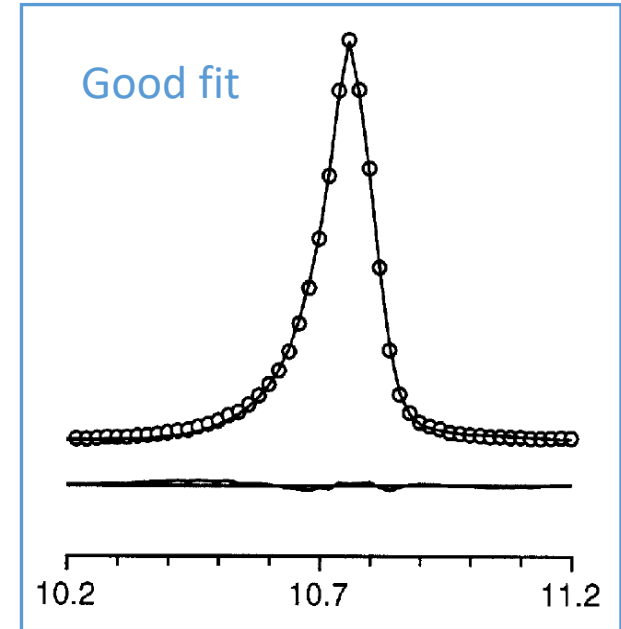
FWHM too small



Peak too symmetrical

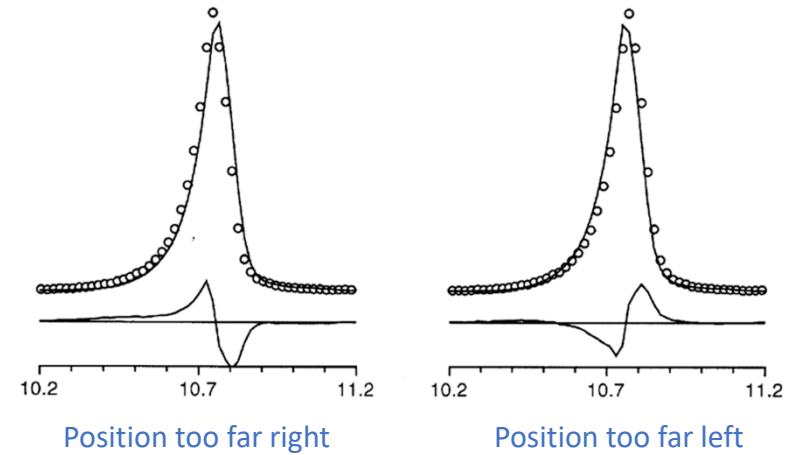
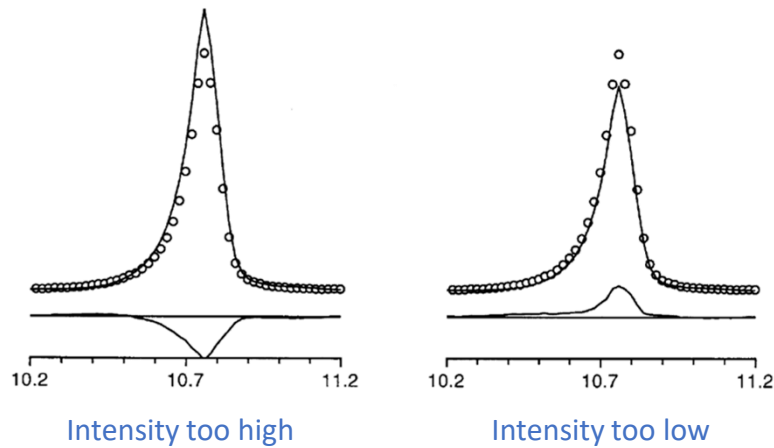


FWHM too small
Asymmetry too small



Good fit

Measured, calculated and difference curves



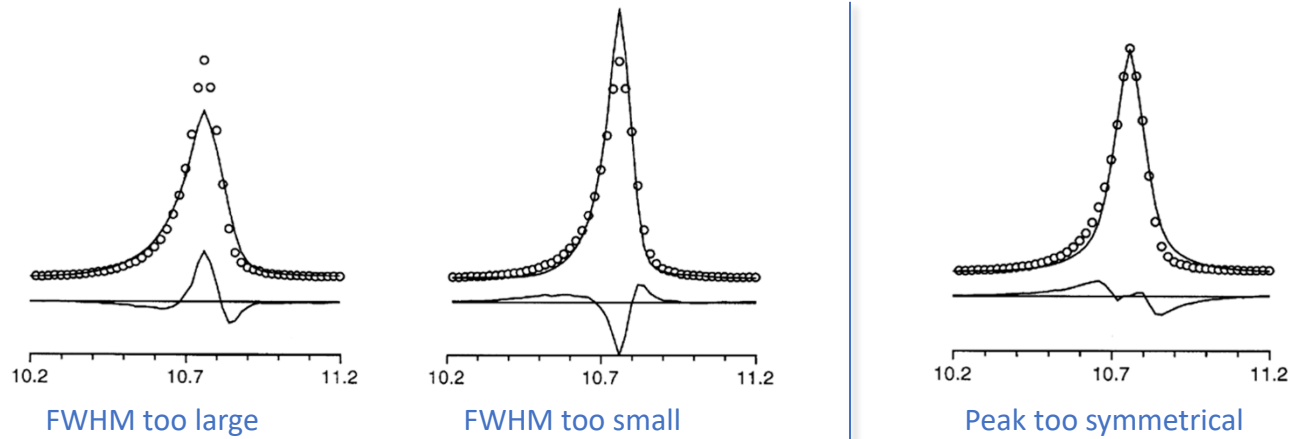
May be caused by:

- Scale factor
- Preferred orientation/ texture/ spottiness
- Atomic model (e.g. atom type/ occupancies)

May be caused by:

- Wrong lattice parameters
- Sample displacement or zero error
- Phase has different symmetry than that used (e.g. tetragonal instead of cubic)

Measured, calculated and difference curves



May be caused by:

- Peak shape model (crystallite size/ strain parameters)
- Anisotropic crystallites and peak shapes

May be caused by:

- Instrumental parameter (e.g. Soller slits)
- Sample transparency

Quality of a fit

$$WSS = \sum_i w_i (I_i^{exp} - I_i^{calc})^2 \rightarrow \min$$

$$w_i = \frac{1}{\sqrt{I_i^{exp}}}$$

$$R_{wp} = \sqrt{\frac{\sum_{i=1}^N [w_i (I_i^{exp} - I_i^{calc})]^2}{\sum_{i=1}^N [w_i I_i^{exp}]^2}}$$

weighted pattern R – value

$$R_{exp} = \sqrt{\frac{N}{\sum_{i=1}^N [w_i I_i^{exp}]^2}}$$

expected R – value

N: number of data points

goodness of fit

$$GofF = \frac{R_{wp}}{R_{exp}}$$

$$\chi^2 = \left(\frac{R_{wp}}{R_{exp}} \right)^2$$

R indices

- The **residual factors** or **reliability factors** (R-factors) measure how good the fit is for:
 - the structural model versus the measured intensities
 - the background
 - the peak positions
 - the peak shapes
- The R-factors help us to decide if a refinement is going in the right direction.

How to use R-factors?

Weighted profile factor – R_{wp}

Should become smaller during a refinement

Tells us if a refinement is going in the right direction

Expected profile factor – R_{exp}

Is the best R_{wp} we can get

Tells us something about counting statistics

Chi-squared χ^2

Is the ratio of $(R_{wp}/R_{exp})^2$

A perfect fit has a $\chi^2 = 1$

How NOT to use R-factors?

Weighted profile factor – R_{wp}

Aim to get below an arbitrary value

Note: Patterns with high background often have good R_{wp} !

Expected profile factor – R_{exp}

Only data dependent – hard to misuse

Chi-squared χ^2

Introduce more and more parameters to get a value $\chi^2 \approx 1$

Compare χ^2 for data from *different instruments* and think that the refinement with the smaller χ^2 is necessarily better

